

Faculdade de Engenharia da Universidade do Porto



CV Report

Task #2-3

8-Ball Pool Table Analysis

2025/2026 – Master in Informatics and Computing Engineering

Authors:

Eduardo Portugal up202206628@up.pt

Pedro Marinho up202206854@up.pt

Xavier Martins up202206632@up.pt

Task 2 Methodology

Dataset Construction

A single parsing stage merges the official main dataset with several auxiliary Roboflow datasets and a “custom” image collection taken by us into one unified json file. For each COCO-formatted source, ball annotations are counted per image, the table-segmentation category is excluded, and a small set of per-dataset correction rules re-weights annotations that represent a fixed group of balls rather than a single ball (e.g. a “break” label is counted as nine balls instead of one). One auxiliary source is additionally filtered to drop a handful of mislabelled filenames and capped at sixteen balls, the physical maximum for an 8-ball table.

Splitting follows two distinct pipelines. For the main dataset, the given partition.csv file is used. For every auxiliary and custom source, we apply a guarantee having ~15% of the total image pool taking part of valid and test sets, none of them contribute to the test set, which stays reserved exclusively for main-dataset images.

Preprocessing

Before counting, every catalogued image is passed through a table-localisation step: a YOLO detector trained separately for table detection proposes a bounding box per image, and the highest-confidence box is used to crop the photograph down to the table surface, with the objective of removing background clutter (players, rails, surrounding furniture).



Model & Training

Ball counting is treated as a 17-way image classification problem (0 to 16 balls) rather than as direct regression. Both formulations were implemented and trained under identical conditions. Classification gave noticeably better metrics, so it was adopted for the final model.

Several models, ranging from a small custom CNN baseline to larger established architectures, were screened with an identical lightweight classification head (256-unit hidden layer, ReLU, dropout 0.3). ConvNeXt-Tiny was selected for the final run after showing the highest accuracy. Mixed-resolution inputs are square-padded to preserve aspect ratio and resized to 300×300, while training augmentation relies on horizontal/vertical flips, $\pm 25^\circ$ rotation, and light color jitter.

Results

Models are compared on the held-out test set using MAE, RMSE, accuracy within ± 1 ball (Acc@1) and exact-match accuracy, then ranked by exact accuracy with RMSE as a tie-breaker.

model	datasets_used	MAE	MSE	RMSE	Acc@1	ExactAcc
ConvNeXt_Tiny	main_dataset,extra-3,custom,extra-6	0.55	0.65	0.80623	0.95	0.5

Datasets used: [main_dataset](#) , [extra-3](#) , “custom” our images, [extra-6](#)

Task 3 Methodology

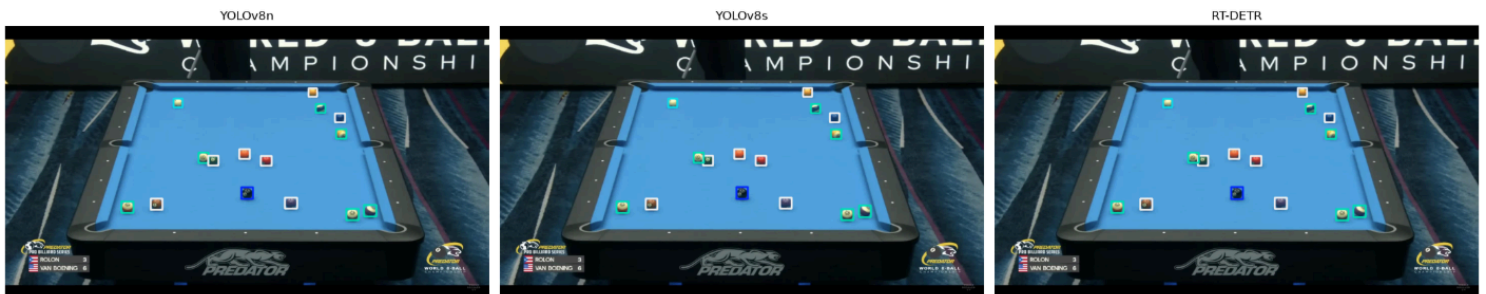
Object Detection

The objective of this task was to detect all pool balls present in an image and classify them into four categories: **Black**, **Cue**, **Solid**, and **Striped**. To evaluate different object detection approaches, three models were compared: **YOLOv8n**, **YOLOv8s**, and **RT-DETR**.

The provided **8-Ball Pool** dataset was used for training and evaluation. Since the original annotations included a **Dot** class corresponding to table markers rather than balls, this class was removed during preprocessing and the remaining labels were remapped accordingly. The official training, validation and test splits of the dataset were preserved throughout the experiments.

All models were initialized from pretrained weights and trained using identical experimental conditions, with an input image size of **640 × 640 pixels**, **50 training epochs**, and **early stopping** (patience = 10). Automatic Mixed Precision (AMP) was enabled to reduce training time and GPU memory usage. The only difference between models was the batch size, which was reduced for RT-DETR due to its higher computational requirements.

The trained models were evaluated on the test set using **Precision**, **Recall**, **mAP@50**, and **mAP@50-95**. In addition to the quantitative evaluation, each model generated predictions containing the detected class, confidence score and bounding box coordinates, which were exported in JSON format to support qualitative analysis and the subsequent table retrieval task.



The best model was RT-DETR, with slightly better mAP@50-95 for all the classification types.

Similar Games

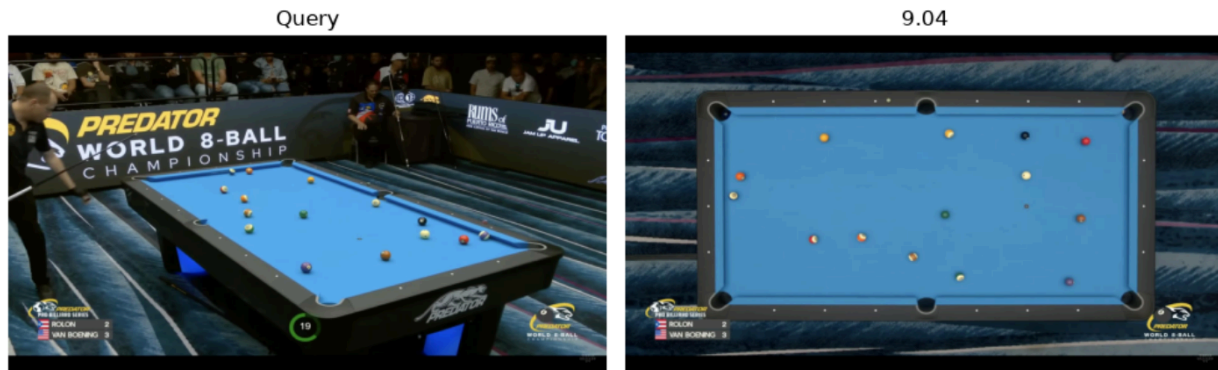
The objective of this task was not only to detect and classify the pool balls, but also to compare different pool table images and identify those representing the most similar game states.

Rather than relying solely on generic feature extraction methods, which would lack contextual awareness, we built upon the results obtained in the previous stage of the project and used them as a fundamental component of the comparison process. Our approach was based on the idea that the classification of the balls would provide the most meaningful representation of a pool game.

First, all balls were detected and classified. Next, the pool table itself was identified through segmentation, allowing us to extract its corners and compute a homography transformation. This process was trained on an [extra dataset](#), specific for segmentation, not used in other parts of the project. The detected ball positions were then projected onto a normalized table reference frame. As a result, the classification step from the previous phase not only identified each ball but also provided its precise position relative to the table after normalization. Because it isn't

certain that all the tables will be viewed from the same side we consider all 4 possible rotations of the table, ensuring a front and back views of the table can be matched together.

With all tables represented in the same coordinate system, the comparison task became significantly more robust. We measured the similarity between different game states by matching corresponding balls and evaluating the distances between their normalized positions. An aggregation strategy was then applied to compute an overall similarity score, and the table with the highest similarity was selected as the best match.



Conclusion

Applying CNN-based models is significantly more robust than relying on simple hand-crafted filtering techniques, largely due to the availability of powerful pre-trained models. However, their application is far from straightforward. One of the main challenges is the difficulty of interpreting the model's decisions and understanding why certain results are obtained. In some cases, this lack of interpretability can make it difficult(or even impossible)to fully explain the model's behavior.

Another major challenge is the creation and collection of relevant, high-quality datasets. The performance of deep learning models is highly dependent on the quality, diversity, and representativeness of the training data. Alternative approaches, such as the artificial generation of datasets through 3D rendering software, were considered as a way to increase data availability and diversity. However, due to time constraints, these approaches could not be explored in depth.

Furthermore, model performance is affected by several factors that are difficult to control, such as dataset biases, lighting variations, occlusions, and differences between training and real-world conditions. Addressing these challenges often requires more advanced models, which can improve accuracy and robustness but also demand significantly greater computational resources, including powerful GPUs and longer training times, beyond those available for this project.

Despite these challenges, this work reinforces the idea that the true potential of computer vision(and artificial intelligence more broadly) is realized when machine learning capabilities are combined with human intelligence, rather than replacing human expertise. The results obtained throughout this project demonstrate how the collaboration between automated learning systems and human knowledge can lead to more robust, efficient, and meaningful solutions.